



Space Propulsion and Mobility

Ushering in a New Era in
Space

A rocket is shown in the process of launching, ascending vertically from a rural landscape. The rocket is white with a dark nose cone and is surrounded by a large plume of white smoke and a bright orange flame at its base. The landscape below is a patchwork of green and brown fields, with a small cluster of buildings visible near the launch site. The sky is a deep blue with scattered white clouds.

Rocket Propulsion II

Design Challenge

Changelog

Version	Date	Comment	Author
1	23.04.2026	First version as shown in first lecture	Chiara Manfletti Alexander Polidar
2	29.04.2026	Specified expander cycle to be closed expander cycle	Alexander Polidar

Setting the Scene

The Chair of Space Mobility and Propulsion conducts experimental research on liquid-propellant rocket engines and their components at the **SPARK test site Oberpfaffenhofen**. The current research focuses on injection & cooling systems and applications of additive manufacturing in those subsystems.

Most current development programs of liquid-oxygen/liquid-methane rocket engines employ **swirl injectors** for the combustion of liquid oxygen with gaseous methane in the main combustion chamber. An alternative injection concept investigated at the Chair is the concept of **porous injectors**. Besides reduced injection pressure drops required, porous injectors promise a short combustion length. This offers the potential to reduce the combustion chamber length required in **expander cycles**.

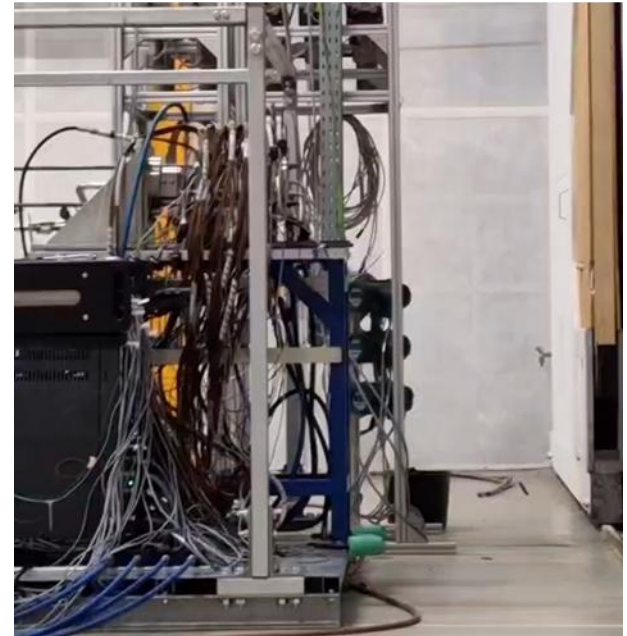
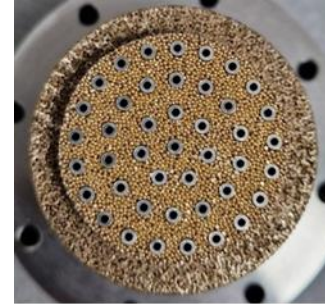


Setting the Scene

Your challenge: Work on different subsystem groups to design, in two teams, baseline engine designs that could be tested at the SPARK test site.

Team 1 shall design a gas generator engine with swirl injectors.

Team 2 shall design an expander engine with porous injectors.



The Challenge



For your design you will have to deliver technically sound and outstanding approaches for:

1. Engine scheme
2. Detailed engine component designs
3. Engine CAD Design

By providing technical documentation explaining your design and justifying any design assumptions made in response to the requirements set.

It is a Design Challenge....so....

....of course there are prizes....

1st prize: Best technical concept

2nd Prize: Most innovative concept that works

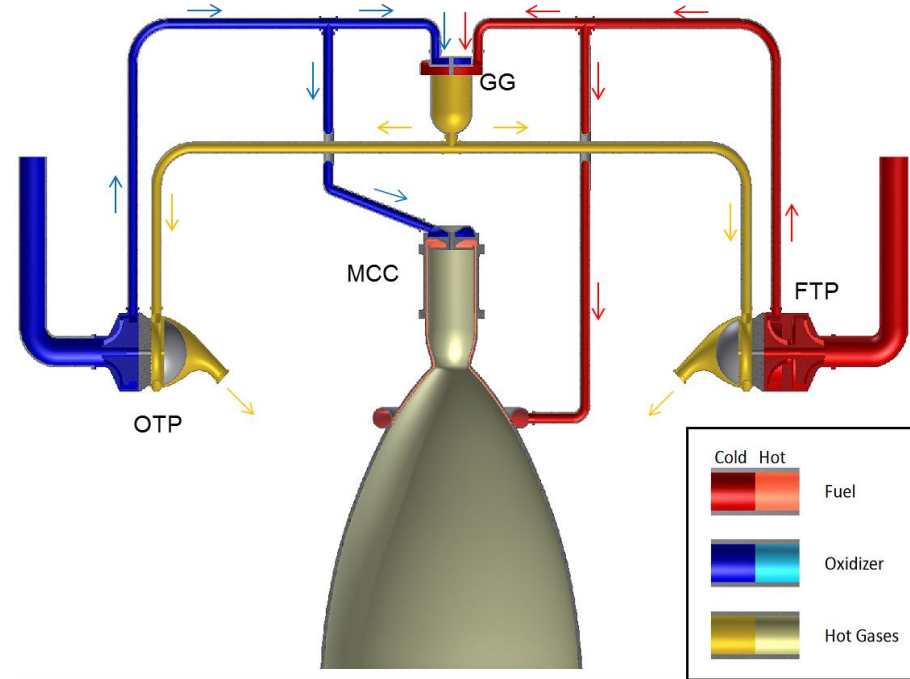
3rd Prize: Team Spirit

- You will be working in groups of 4
- Each group will be focusing on either the engine system as a whole or 1 engine subsystem assembly from the available list of engine assemblies
- The engine components will be assigned to groups on a first-come-first-served basis
- Between groups you will be competing for prizes but also working together 😊
- Each group will be providing output parameters of their assemblies to the other groups

Gas Generator Cycle

Subsystems

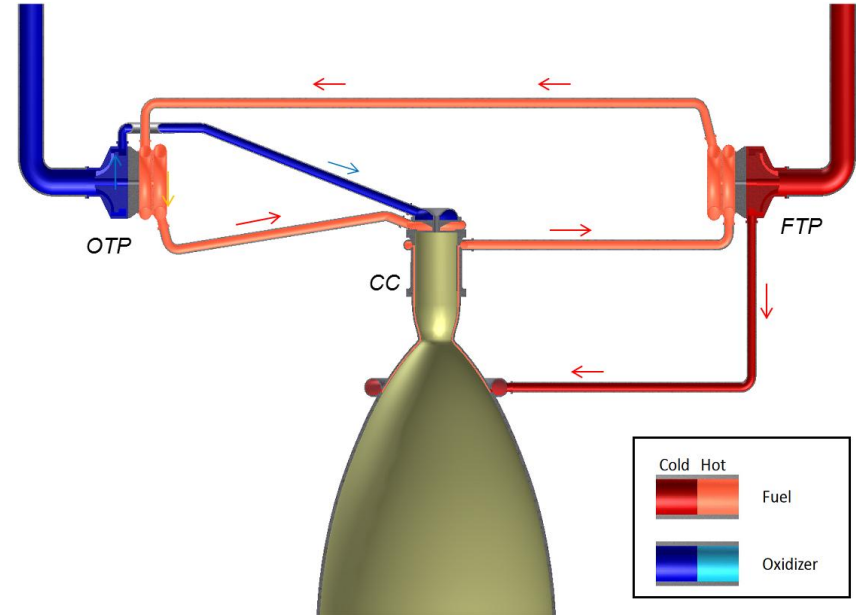
- Power pack
 - **Turbopump** (single or two shafts)
 - Gas generator
- **Thrust chamber** (incl. nozzle)
- **Injector head**
- Feed & control system
- Ignition system



Expander Cycle

Subsystems

- **Turbopump** (single or two shafts)
- **Thrust chamber** (incl. nozzle)
- **Injector head**
- Feed & control system
- Ignition system



List of Engine Assemblies

- **A0: Engine System**
- **A1: Turbomachinery**
- **A2: Thrust Chamber** incl. Cooling System
- **A3: Injector Head** incl. Injection Manifolds and Injectors

If more than 4 groups are formed per team, the following subsystems can be selected:

- **A4: Feed and Control System** incl. Valves
- **A5: Ignition System**
- **A6: Gas Generator**
- **A7: Thrust Vector Control System**

In addition to overall System and Engine requirements, each Assembly, A1-A7, has a set of specific requirements it must fulfil.

Engine Requirements

Engine Requirements

Nr.	Requirement
E-R01	The Engine shall be a liquid bi-propellant gas generator (Team 1) or closed expander (Team 2) engine
E-R02	The propellant combination shall be LOx/LCH4
E-R03	The Engine shall provide 40 kN of sea-level Thrust
E-R04	The nominal combustion chamber pressure shall be 80 bar
E-R05	LOx shall be supplied by external run tanks at 2 bar minimum and 90 K in the LOx tank
E-R06	LCH4 shall be supplied by external run tanks at 2 bar minimum in the LCH4 tank
E-R07	The subsystems shall be designed with 10% margin

Power Pack Requirements



Nr.	Requirement
PP-R01	The power pack shall minimize efficiency losses
PP-R02	The power pack shall be re-ignitable
PP-R03	The power pack shall be easily manufacturable

Thrust Chamber Assembly Requirements



Nr.	Requirement
CC-R01	The Combustion Chamber shall be easily manufacturable
CC-R02	The Combustion Chamber design shall consider an integral design printable with 3D printing methods
CC-R03	The Combustion Chamber shall be cooled regeneratively.

Injector Head Requirements



Nr.	Requirement
Inj-R01	The Injector Head technology shall be swirled injection (Team 1) or porous injection (Team 2)

Feed and Control System Requirements



Nr.	Requirement
FCS-R01	The feed system shall provide the required mass flows at the required pressure from the pumps to the injector head and the cooling system
FCS-R02	The feed and control system shall allow to monitor the operation of the engine
FCS-R03	The feed system shall be designed to comply with operating pressures and temperatures

Ignition System Requirements

Nr.	Requirement
Ign-R01	The ignition system shall be reusable
Ign-R02	The ignition system shall not rely on toxic propellants
Ign-R03	The ignition sequence shall be tuned such that overpressure during start up is prevented

Thrust Vector Control Requirements

Nr.	Requirement
TVC-R01	The Thrust Vector Control shall allow +/- 5° deflection in each direction
TVC-R02	The Thrust Vector Control shall not require more than 1s to transition to a new position
TVC-R03	The system state (deflection) shall be tracked by a controller (Either a controller is directly provided with a COTS actor, or the position of the actor can be measured and fed back as analog / digital signal; it is NOT part of this project to synthesize the control law)

Expectation Horizons

Leading question: How does the injector head concept and its design influence the overall engine cycle if it's 1) a gas generator cycle, 2) an expander cycle?

The focus is on:

- Injection system
- Thrust chamber regenerative cooling
- Turbopumps

The engine system groups of Team 1 and Team 2 shall coordinate the overall design, manage the interfaces and identify together with their subsystem groups the interplay of:

- **Injection system pressure drop vs. turbomachinery design:** How sensitive is your engine cycle to variations in the injection system pressure drop?
- **Wall heat load profile vs. cooling system design:** How sensitive is your engine cycle to variations in the wall heat flux profile?

Expectation Horizon: A00 – Engine System

The expected scope contains the following tasks, but the outcome must not be limited to it. A more detailed design is always encouraged if you manage within the given time.

- Exchange closely with subsystem teams and trigger exchange between teams
- Conduct engine system trade-offs in exchange with subsystem teams
→ requires identification of dependencies
- Define common interfaces and ensure that subsystems use matching interfaces
- Create an engine mass budget & efficiency budget (see next slide)
- Validate that the engine system requirements are met
- Create an overall assembly engine in CAD

Expectation Horizon: A00 – Engine System

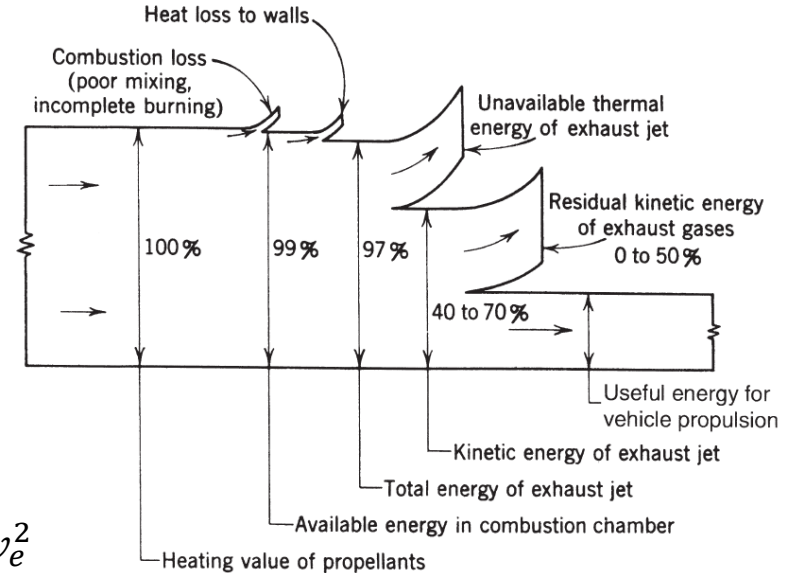
Efficiency budget

Internal efficiency of a propulsion system

- fig. on the right: only combustion chamber
- to be added by you: Losses in other components

$$\eta_{int} = \frac{\text{kinetic power (jet)}}{\text{available chemical power (propellants)}} = \frac{\frac{1}{2} \dot{m} v_e^2}{P_{chem}}$$

with $P_{chem} = \dot{m} Q_{LHV} \rightarrow Q_{LHV}$: lower heating value of fuel /heat of combustion without condensation of evaporated water in the combustion products



Source: [Sutton2010] on Moodle

Expectation Horizon: A01 – Turbomachinery



The expected scope contains the following tasks, but the outcome must not be limited to it. A more detailed design is always encouraged if you manage within the given time.

Note that „analytical methods“ don't exclude to develop design codes or applying numerical solution methods, but it emphasizes that CFD or FEM simulations are not mandatory.

- Evaluate and select single-shaft vs. two-shaft configuration
- Preliminarily design a single pump-turbine configuration based on analytical methods
- Select construction solutions, e.g. bearing type, based on qualitative (e.g., morphological box) or quantitative analyses
- Conduct trade-off studies on the design parameters; optional: apply optimization methods
- Create CAD design of single components and the assembly
- Create an assembly drawing and an exploded view of the assembly

Expectation Horizon: A02 – Thrust Chamber



The expected scope contains the following tasks, but the outcome must not be limited to it. A more detailed design is always encouraged if you manage within the given time.

Note that „analytical methods“ don't exclude to develop design codes or applying numerical solution methods, but it emphasizes that CFD or FEM simulations are not mandatory.

- Define the main geometrical parameters (e.g., L^* , A_t , ε) in accordance with the propellants and the injection system chosen
- Design the chamber contour
- Design the cooling system based under consideration of the axial wall heat flux profile and thermal and structural limits
- Select materials and manufacturing methods
- Create CAD design of single components and the assembly
- Create an assembly drawing and an exploded view of the assembly

Expectation Horizon: A03 – Injector Head



The expected scope contains the following tasks, but the outcome must not be limited to it. A more detailed design is always encouraged if you manage within the given time.

Note that „analytical methods“ don't exclude to develop design codes or applying numerical solution methods, but it emphasizes that CFD or FEM simulations are not mandatory.

- Create a preliminary design to derive the main performance and geometrical parameters of the injection system
- Create CAD design of single components (injection elements, manifolds, ...) and the assembly
- Create an assembly drawing and an exploded view of the assembly

Expectation Horizon: A04 – Feed and Control System



The expected scope contains the following tasks, but the outcome must not be limited to it. A more detailed design is always encouraged if you manage within the given time.

Note that „analytical methods“ don't exclude to develop design codes or applying numerical solution methods, but it emphasizes that CFD or FEM simulations are not mandatory.

- Create a Piping and Instrumentation Diagram (P&ID) that shows all valves, other fluid components and sensors of the engine
- Describe the ignition and shut-off sequence of your engine step-by-step by referring to the P&ID
- Select valve types based on their function required
- Create CAD design that features all pipes, coarse geometries of valves and other fluid components
- Create assembly drawings

MILESTONES

Milestones



M1 – List of components & subsystem interfaces –
Deadline: 21.05.2026



M2 – Component parameter freeze –
Deadline: 02.07.2026



M3 – Interface specifications and overall dimensions –
Deadline: 16.07.2026



M4 – CAD incl. drawings & assembly description –
Deadline: Report submission

M1 – List of components & subsystem interfaces



By this deadline, you have to deliver a component overview and a list of interfaces, which your subsystem shares with other subsystems:

Component list: Which components does your system consist of?

Interfaces: Fill Excel „interface document“ with

- Fluidic interfaces: Inlet/outlet fluid properties of components (mass flow, pressure, temperature, density)
- Physical interfaces: Pipe connection, joints, bolts etc.

Specifications of the interfaces are not required until this deadline.

Submission: Filled Excel „interface document“ → Review in exercise with teaching assistant at the deadline
→ sheets „Component Overview“, „Fluidic Interfaces“, „Physical Interfaces“

M2 – Component parameter freeze

By this deadline, you have to deliver the main parameters that describe your subassembly, e.g.:

- Mass flow(s)
- Pressure(s)
- Mass budget estimation(s)

But also subassembly parameters which are required by other subsystems

→ Communicate to other teams if you require parameters from them

→ Check that your values match with interfacing subassemblies and overall engine parameters

You can make changes to your subassembly afterwards, but these won't have to be considered by other subassemblies

Submission: Filled Excel „interface document“ → Review in exercise with teaching assistant at the deadline
→ sheets „Fluidic interfaces“, „Mass budget“

M3 – Interface specifications and overall dimensions



By this deadline, you have to deliver the physical interface definition of interfaces that you have with other subassemblies, e.g.:

- Fluid interfaces
- Phsysical interfaces
- ...

And the overall dimensions of your subassembly components, e.g.:

- Cylindrical components: Length, diameter
- Rectangular components: Length, width, height

The physical interfaces must not be changed afterwards to be compatible with other teams

Submission: Filled Excel „interface document“ → Review in exercise with teaching assistant at the deadline
→ sheets „Fluidic interfaces“, „Physical interfaces“